THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

December 18, 2018 2:30 p.m. – 5:00 p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL SIX questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (30 points)

Prove or disprove each of the following statements:

(a) Let

$$\begin{aligned} A &= \{a \in \mathbb{R} : a = p + q\sqrt{2} \text{ for some } p, q \in \mathbb{Q}\}, \\ B &= \{b \in \mathbb{R} : b = r + s\sqrt{3} \text{ for some } r, s \in \mathbb{Q}\}. \end{aligned}$$

Then $A \cap B = \mathbb{Q}$.

(*Hint*: You may use the fact that $\sqrt{2}$, $\sqrt{3}$ and $\sqrt{6}$ are irrational.)

(b) For any nonempty sets A, B and for any function $f : A \longrightarrow B$,

$$f(S) \setminus f(T) = f(S \setminus T) \quad \text{for any subsets } S \text{ and } T \text{ of } A.$$

(c) For any nonempty sets A, B and for any function $f : A \longrightarrow B$,

$$f^{-1}(U) \setminus f^{-1}(V) = f^{-1}(U \setminus V) \quad \text{for any subsets } U \text{ and } V \text{ of } B.$$

(d) There exist uncountable sets A and B such that $A \cap B$ is denumerable.

(e) For any denumerable set A, there exists a function from A to $\mathbb{N}$ that is surjective but not injective.

2. (12 points)

Consider the sequence $F_1, F_2, F_3, \ldots$ of *Fibonacci numbers* satisfying $F_1 = 1$, $F_2 = 1$ and

$$F_n = F_{n-1} + F_{n-2} \quad \text{for any integer } n \geq 3.$$

(a) Prove by induction that for any integer $n \geq 2$,

$$F_n^2 - F_{n+1}F_{n-1} = (-1)^{n+1}.$$

(b) Use (a) to prove that $5F_n^2 + 4$ or $5F_n^2 - 4$ is a perfect square for any $n \in \mathbb{N}$.

3. (12 points)

(a) Prove that for any $m \in \mathbb{Z}$, if m is odd, then $m^2 \equiv 1 \pmod 8$.

(b) Use (a), or otherwise, to show that the only positive integer solution of the equation

$$m^2 + 8 = 3^n$$

is $(m, n) = (1, 2)$.

(*Hint*: You may use the fact that for any $x \in \mathbb{Z}$, x is odd if and only if x^2 is odd.)

4. (16 points)

Let A and B be nonempty sets and let $f : A \longrightarrow B$ be a surjective function. For each $b \in B$, define

$$A_b = f^{-1}(\{b\}),$$

the preimage set of $\{b\}$ under f.

(a) Prove that A_b is nonempty for any $b \in B$.

(b) Let $P = \{A_b : b \in B\}$. Prove that P is a partition of A.

(c) Let $A = \mathbb{Z}$ and $B = \mathbb{Z}_2 \times \mathbb{Z}_3$, and let $f : \mathbb{Z} \longrightarrow \mathbb{Z}_2 \times \mathbb{Z}_3$ be defined by

$$f(a) = ([a+1]_2, [2a]_3),$$

where $[a]_n$ denotes the equivalence class with respect to the congruence of integers modulo n.

(i) Prove that f is surjective.

(ii) Prove that $f^{-1}(\{([0]_2, [0]_3)\}) = [3]_6$. Write down (without proof) the preimage set $f^{-1}(\{([b]_2, [c]_3)\})$ for all other $([b]_2, [c]_3) \in \mathbb{Z}_2 \times \mathbb{Z}_3$.

(iii) Use the above results to construct a partition of $\mathbb{Z}$.

5. (15 points)

For any nonempty sets A, B, let $f : \mathcal{P}(A \cup B) \longrightarrow \mathcal{P}(A) \times \mathcal{P}(B)$ be defined by

$$f(X) = (X \cap A, X \cap B).$$

(a) Prove that f is injective.

(b) Show, by giving an example of A and B, that f is not surjective.

(c) The function f will become surjective if an assumption is added on the sets A and B. What is this assumption? Prove that f is surjective under this additional assumption.

6. (15 points)

Let A be a nonempty set and let $f : A \longrightarrow A$ be a bijective function. For any $n \in \mathbb{N}$, denote by f^n the composition of n times of f (which is also a bijective function), that is,

$$f^n = \overbrace{f \circ f \circ \cdots \circ f}^{n \text{ times}}.$$

Define $f^0 = id_A$, the identity function of A. Define f^{-n} for any $n \in \mathbb{N}$ to be $(f^{-1})^n$, where f^{-1} is the inverse of f.

(a) Prove that $f^m \circ f^n = f^{m+n}$ for any $m, n \in \mathbb{Z}$.

(b) Prove that $(f^m)^{-1} = f^{-m}$ for any $m \in \mathbb{Z}$.

(c) Let R be the relation on A defined by xRy if $y = f^m(x)$ for some $m \in \mathbb{Z}$. Prove that R is an equivalence relation on A.

******* **END OF PAPER** *******